Sri Chaitanya Educational Institutions | Infinity Learn

JEE Advance Grand Test - 8

Test Instructions

1. Total duration of this test is **180** minutes.
2. This test has 3 subjects consisting of **54** questions in total.
3. There are **9** total sections in the test.
4. Sections Info :

Physics

a. **Section A** has **7** questions, compulsory questions **7**. **3** marks will be given for correct attempt and incorrect attempt **-1** .
b. **Section B** has **7** questions, compulsory questions **7** and is eligible for **partial marking** .
   i. **4 marks** will be given if all correct options are chosen.
   ii. **3 marks** will be given if all the four options are correct but ONLY three correct options are chosen.
   iii. **2 marks** will be given if all the four options are correct but ONLY two correct options are chosen.
   iv. **1 marks** will be given if all the four options are correct but ONLY one correct option is chosen.
   v. **2 marks** will be deducted if any incorrect option is chosen.
c. **Section C** has **4** questions, compulsory questions **4**. **3** marks will be given for correct attempt and incorrect attempt **0** .

Chemistry

a. **Section A** has **7** questions, compulsory questions **7**. **3** marks will be given for correct attempt and incorrect attempt **-1** .
b. **Section B** has **7** questions, compulsory questions **7** and is eligible for **partial marking** .
   i. **4 marks** will be given if all correct options are chosen.
   ii. **3 marks** will be given if all the four options are correct but ONLY three correct options are chosen.
   iii. **2 marks** will be given if all the four options are correct but ONLY two correct options are chosen.
   iv. **1 marks** will be given if all the four options are correct but ONLY one correct option is chosen.
   v. **2 marks** will be deducted if any incorrect option is chosen.
c. **Section C** has **4** questions, compulsory questions **4**. **3** marks will be given for correct attempt and incorrect attempt **0** .

Mathematics

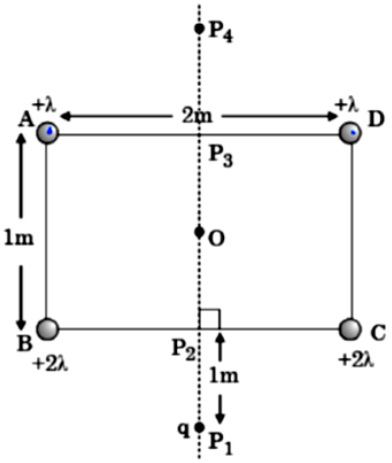
a. **Section A** has **7** questions, compulsory questions **7**. **3** marks will be given for correct attempt and incorrect attempt **-1** .
b. **Section B** has **7** questions, compulsory questions **7** and is eligible for **partial marking** .
   i. **4 marks** will be given if all correct options are chosen.
   ii. **3 marks** will be given if all the four options are correct but ONLY three correct options are chosen.
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   iv. **1 marks** will be given if all the four options are correct but ONLY one correct option is chosen.
   v. **2 marks** will be deducted if any incorrect option is chosen.
c. **Section C** has **4** questions, compulsory questions **4**. **3** marks will be given for correct attempt and incorrect attempt **0** .

5. Total marks for this test is **183** marks.
6. No marks will be deducted for unattempted questions.
7. This test can be submitted only once.
8. Once the test has been submitted, you cannot edit the responses.
9. Results will be anounced post test submission.
10. The test will be auto-submitted once the timer ends.

Physics

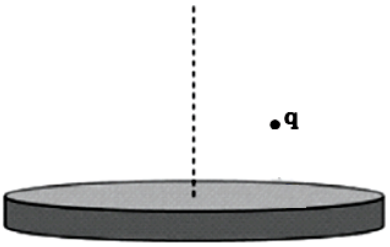
Section A

1. There is a system of 4 infinite length uniformly charged wire passing through corners of a rectangle ABCD and perpendicular to the plane of rectangle as shown in figure. Charge density of wires passing through A, D is λ and B and C is 2λ. P_1, P_2, P_3 and P_4 are points on the perpendicular bisector of AD and BC. (q is positive charge)



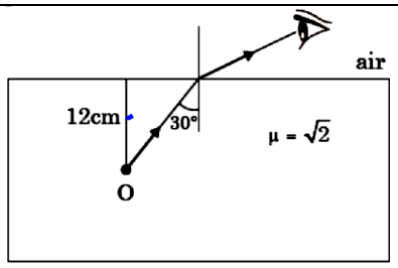
1. Potential energy of system is increasing continuously if a charge particle 'q' is moved from P_1 to O along perpendicular bisectors
2. Potential energy of system 1st increase then decrease if a charge particle 'q' is moved from O to P_3
3. Potential energy of system decrease continuously if a charge particle 'q' is moved from P_2 to O along perpendicular bisector line
4. Potential energy of system first increases and then decreases if charge is moved from P_2 to O along perpendicular bisector line.

2. A light disc made of aluminium (a nonmagnetic material) is kept horizontally and is free to rotate about its axis as shown in the figure. A point charge particle is revolving about the axis of disc, then the disc will



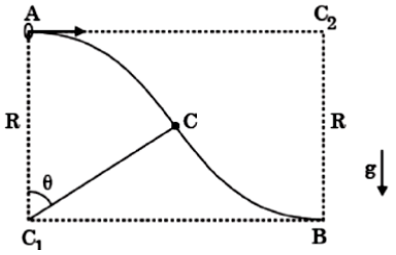
1. Rotate in the direction opposite to the direction of charge's motion
2. Rotate in same direction as the direction of charge's motion.
3. Not rotate but temperature will increase
4. Not rotate but its temperature will remain unchanged

3. An object is placed at distance 12 cm from top face of glass slab $\mu = \sqrt{2}$. If the object is viewed from the position of observer as shown, then the apparent distance from the top surface of glass slab will be:



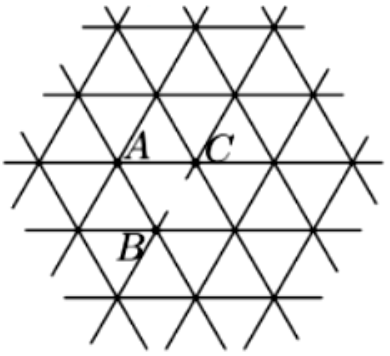
1. $\frac{4}{\sqrt{3}}$
2. $4\sqrt{3}$
3. $2\sqrt{2}$
4. $\frac{8}{\sqrt{3}}$

4. Smooth wire shaped as circular arc AC and CB are of same radius but centre of curvature are on opposite side as shown. A small bead is projected horizontally from A with speed such that normal force become zero just before point C. Speed of the particle when it reaches to B will be [Take $\theta = \cos^{-1}\left(\frac{5}{6}\right)$]



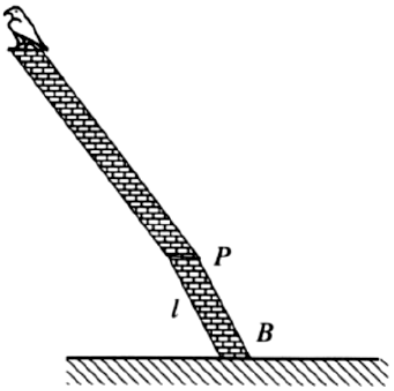
1. $\sqrt{2gR}$
2. $\sqrt{3gR}$
3. $\sqrt{\frac{5}{2}gR}$
4. $\sqrt{\frac{3}{2}gR}$

5. There is an infinite triangular lattice; the edges of the lattice are made of wire, and the resistance of each edge is R. Let us denote the corners of a triangular lattice face by A, B and C. The wire connecting B and C is cut off. Determine the resistance between A and B.



1. $\frac{R}{4}$
2. $\frac{R}{6}$
3. $\frac{R}{8}$
4. $\frac{3R}{8}$

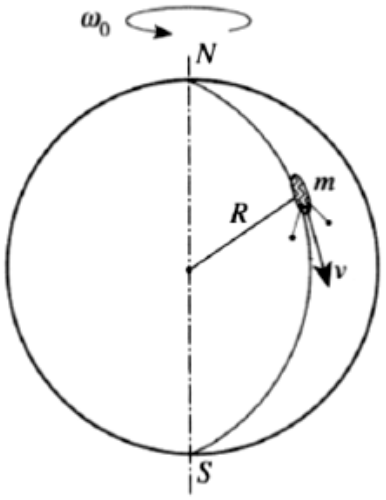
6. A tall, slender, cylindrical brick chimney of height L is slightly perturbed from its vertical equilibrium position so that it topples over, rotating rigidly around its base B until it breaks at a point P. Show that the most likely value for the distance of P from B is L/n. Assume that the chimney breaks because the torque is too great and the chimney bends and snaps



Value of n is

1. 4
2. 2
3. 3
4. 6

7. A toy globe rotates freely without friction with an initial angular velocity ω_0. A bug starting at one pole N travel to the other pole S along a meridian with constant velocity v. The axis of rotation of the globe is held fixed. Let M and R denote the mass and radius of the globe (a solid sphere, moment of inertia $I_0 = 2MR^2/5$), m the mass of the bug, and T the duration of the bug's journey.

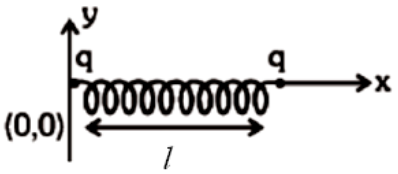


Show that, during the time the bug is travelling, the globe rotates through an angle.

1. $\Delta\theta = \frac{\pi\omega_0 R}{v}\sqrt{\frac{2M}{(2M+5m)}}$
2. $\Delta\theta = \frac{2\pi\omega_0 R}{v}\sqrt{\frac{2M}{2M+5m}}$
3. $\Delta\theta = \frac{\pi\omega_0 R}{v}\sqrt{\frac{2M}{2M+7m}}$
4. $\Delta\theta = \frac{2\pi\omega_0 R}{v}\sqrt{\frac{2M}{2M+7m}}$

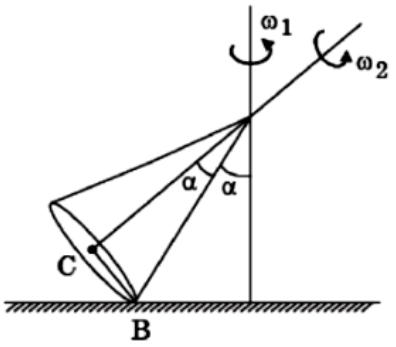
Section B

8. One end of a spring of negligible unstretched length and spring constant k is fixed at the origin (0, 0). A particle of mass m carrying a positive charge q is attached at its other end. The entire system is kept on a smooth horizontal surface. When some positive point charge ‘q’ is fixed at the origin, the spring get stretched to a length l and attains a new equilibrium position and at rest. If the spring is elongated slightly from equilibrium and released, it is found that



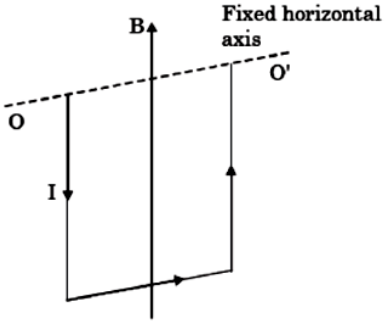
1. Particle oscillate with time period $T = 2\pi\sqrt{\frac{m}{3k}}$
2. Particle oscillate with time period $T = \pi\sqrt{\frac{m}{3k}}$
3. When particle move from mean position to extreme position then modulus of change in potential energy of spring is equal to modulus of change in electrostatic potential energy (for very small amplitude)
4. When particle move from mean position to extreme position then modulus of change in potential energy of spring is half modulus of change in electrostatic potential energy (for very small amplitude).

9. A uniform solid cone (height h, semi–vertex angle α) is inclined against a vertical axis as shown in the figure. The cone rotates about its own axis as well as rotates about the vertical axis with angular speeds marked in the diagram. If the cone does not slip at point B, then



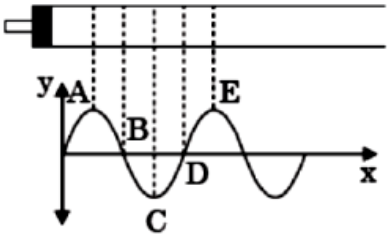
1. $\frac{\omega_1}{\omega_2} = 1$
2. $\frac{\omega_1}{\omega_2} = \sqrt{3}$
3. Velocity of $C = h\omega_1 \sin 2\alpha$
4. Velocity of $C = h\omega_1$

10. A copper wire of density ρ with cross–sectional area S bent to make three sides of a square can turn about a horizontal axis OO'. The wire is located in uniform vertical magnetic field. If on passing a current I through the wire the later deflects by an angle θ and remain in equilibrium. Then



1. Magnetic filed $B = \frac{2\rho Sg}{I}\tan\theta$
2. Magnetic filed $B = \frac{2\rho Sg}{I}\sin\theta$
3. Net magnetic force on the copper wire is along horizontal
4. Net magnetic force on the copper wire is zero

11. Displacement (y) of air column of position x from its mean position at any instant is given by graph shown in the figure. We can conclude from graph that

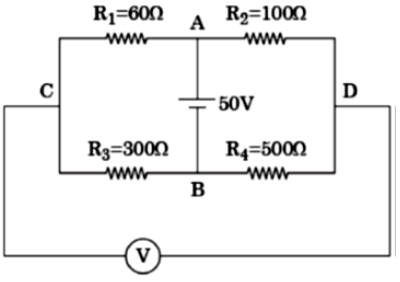


1. Average density in pipe is equal to density at point A at given instant
2. Average density in pipe is equal to density at point C at given instant
3. Density of air is maximum at B
4. Density of air is minimum at D

12. A proton collides with a stationary deuteron and a ^3He nucleus is formed. For this reaction to take place, the proton must have minimum kinetic energy 1.4 MeV. If instead, a deuteron collides with a stationary proton to make a ^3He nucleus. Then

1. Minimum kinetic energy deuteron must possess = 2.8 MeV
2. Minimum kinetic energy deuteron must possess = 0.7 MeV
3. Energy needed for the reaction is approx 0.93 MeV
4. Energy needed for the reaction is approx 2.8 MeV

13. In the balanced condition, the values of the resistances of the four arms of a wheatstone bridge are shown in figure below. The resistance R_3 has temperature coefficient $\alpha = KT\,^\circ C^{-1}$ where $K = 0.0008$ and T is temperature in $^\circ C$. If the temperature of only R_3 is increased by $10^\circ C$, then [Take initial temperature $0^\circ C$; Resistance of voltmeter V is very large]

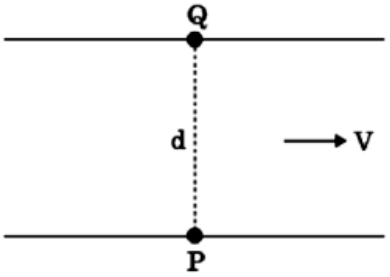


1. The voltage developed between C and D will be approx 0.27 volt at $T = 10^0 C$
2. The voltage developed between C and D will be approx 0.47 volt at $T = 10^0 C$
3. The voltage developed between C and D will be 0 volt at

$T = 0^0C$

4. The voltage developed between C and D will be 0.5 volt at $T = 0^0C$

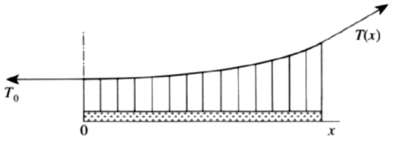
14. A boat starting from point 'P' on one bank of river with a sound siren of frequency f moves in the river in such a way that it crosses the river in minimum time. A detector is fixed at some height from the ground on the other bank just opposite to point P. Velocity of river flow is V, velocity of boat is 2V w.r.t. water flow and velocity of sound in air = V_0 . Consider only sound pulses reaching the detector through air. Then [Width of river = d and $V << V_0$]



1. Maximum frequency observed by detector = $\left[\frac{V_0}{V_0-2V}\right]f$
2. Maximum frequency observed by detector = $\left[\frac{V_0}{V_0-\sqrt{3}V}\right]f$
3. Detector will observed frequency f when boat is at the middle of river
4. Minimum frequency observed by detector = $\left[\frac{V_0}{V_0+3V}\right]f$

Section C

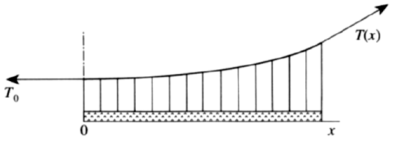
15. *A flexible massless cable in a suspension bridge is subject to uniform loading along the x–axis. The weight of the load per unit length of the cable is ω, and the tension in the cable at the center of the bridge (at x = 0) is* T_0.



Find the shape of the cable at equilibrium.

1. $y = y_0 + \frac{\omega x^2}{2T_0}$
2. $y = y_0 + \frac{\omega x^2}{T_0}$
3. $y = y_0 + \frac{\omega x^2}{4T_0}$
4. $y = y_0 + \frac{2\omega x^2}{T_0}$

16. *A flexible massless cable in a suspension bridge is subject to uniform loading along the x–axis. The weight of the load per unit length of the cable is ω, and the tension in the cable at the center of the bridge (at x = 0) is* T_0.



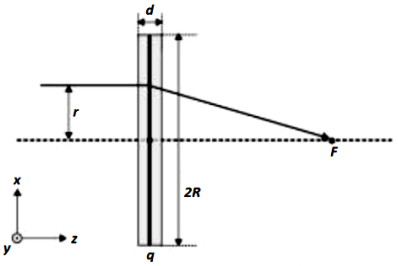
What is the tension T(x) in the cable at position x at equilibrium?

1. $T = T_0\sqrt{1+\left(\frac{\omega x}{2T_0}\right)^2}$
2. $T = T_0\sqrt{1+\left(\frac{\omega x}{T_0}\right)^2}$

3. $T = T_0\sqrt{1+\left(\frac{2\omega x}{T_0}\right)^2}$

4. $T = T_0\sqrt{1+\left(\frac{\omega x}{4T_0}\right)^2}$

17. *One wants to build a device to focus electrons–an electrostatic lens. Let us consider the following construction. The ring is situated perpendicularly to the z–axis, as shown in figure. We have a source that produces on–demand packets of non–relativistic electrons. Kinetic energy of these electrons is* $E = \frac{mv^2}{2}$ *(* v *is very high velocity) and they leave the source at precisely controlled moments. The system is programmed so that the ring is charge–neutral mosts of the time, but its charge becomes q when electrons are closer than a distance* $\frac{d}{2}(d << R)$ *from the plane of the ring (shaded region in figure, called "active region"). In part C assume that charging and de–charging processes are instantaneous and the electric field "fills the space" instantaneously as well. One can neglect magnetic fields and assume that the velocity of electrons in the z–direction is constant. Moving electrons do not perturb the charge distribution on the ring.*
Note: In the immediate vicinity of the opening the field is non–uniform. To estimate the radial component of the electric field in this region, apply Gauss' Law to a short coaxial cylinder passing through the hole.



A model of an electrostatic lens

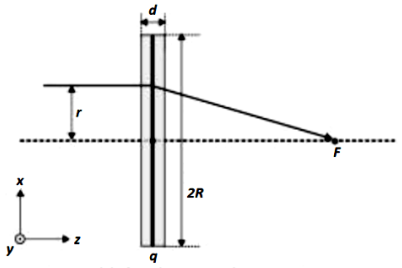
Radial component of electric field E(r) in the active region (r < < R).

1. $\frac{qr}{8\pi\varepsilon_0 R^3}$
2. $\frac{qr}{4\pi\varepsilon_0 R^3}$
3. $\frac{qr}{2\pi\varepsilon_0 R^3}$
4. 0

18. *One wants to build a device to focus electrons–an electrostatic lens. Let us consider the following construction. The ring is situated perpendicularly to the z–axis, as shown in figure. We have a source that produces on–demand packets of non–relativistic electrons. Kinetic energy of these electrons is* $E = \frac{mv^2}{2}$ *(* v *is very high velocity) and they leave the source at precisely controlled moments. The system is programmed so that the ring is charge–neutral mosts of the time, but its charge becomes q when electrons are closer than a distance* $\frac{d}{2}(d << R)$ *from the plane of the ring (shaded region in figure, called "active region"). In part C assume that charging and de–charging processes are instantaneous and the electric field "fills the space" instantaneously as*

well. One can neglect magnetic fields and assume that the velocity of electrons in the z–direction is constant. Moving electrons do not perturb the charge distribution on the ring.

Note: In the immediate vicinity of the opening the field is non–uniform. To estimate the radial component of the electric field in this region, apply Gauss' Law to a short coaxial cylinder passing through the hole.



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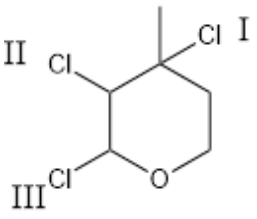
Focal length f of this electrostatic lens for (r < < R).

1. $\frac{8\pi\varepsilon_0 R^3 E}{qed}$
2. $\frac{16\pi\varepsilon_0 R^3 E}{qed}$
3. $\frac{4\pi\varepsilon_0 R^3 E}{qed}$
4. $\frac{32\pi\varepsilon_0 R^3 E}{qed}$

Chemistry

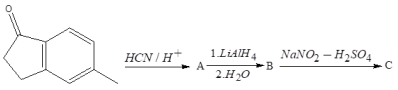
Section A

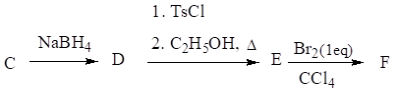
19. The rate of reaction of aqueous silver nitrate, with labeled chlorine atoms, in the following molecule is given by the order:



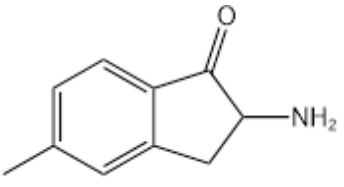
1. I > II > III
2. III > I > II
3. III > II > I
4. I > III > II

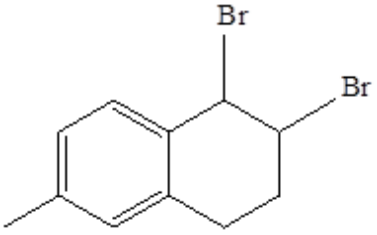
20. In the given reaction sequence, the compound F(major) is

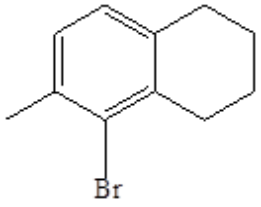


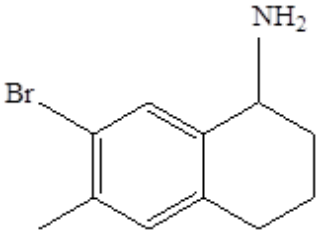


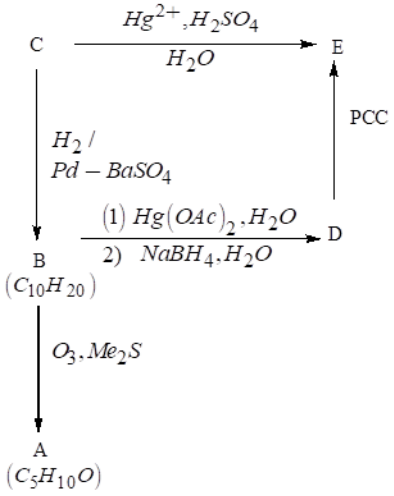
The compound F(major) is.

1. 

2. 

3. 

4. 

21. 

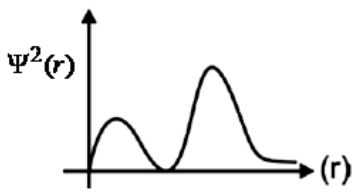
Molecular mass of 'A' can increase maximum by 1 unit when placed in D_3O^+/D_2O for long time. 'A' gives metallic silver when heated with ammonical silver nitrate solution. Number of optically active isomers for C & D are _____ respectively

1. 4, 4
2. 3, 8
3. 2, 8
4. 8, 8

22. Bromine halide (R) $+H_2O \rightarrow X + Y$, [major products] Electronic geometry of R is a distorted octahedron. Choose correct pair of X and Y.

1. X= HBr Y=HOF
2. X=HBrO Y=HF
3. Y= $HBrO_2$ X=HF
4. Y= $HBrO_3$ X=HF

23. A qualitative plot of variation of probability density, $\Psi^2(r)$, as a function of distance(r) of the electron (X) from the nucleus of hydrogen like species of atomic number 'Z' is given below. Which of the given statements is incorrect?



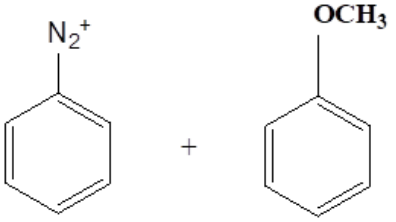
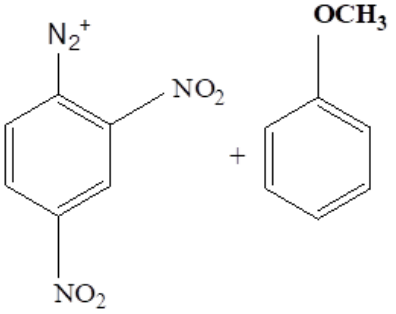
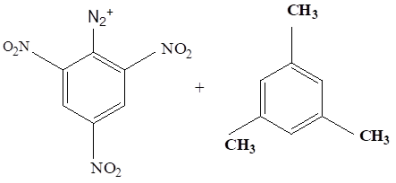
1. The difference between the principal quantum number and the azimuthal quantum number for X is 2
2. Average potential energy of X can be $\approx -3.02 \ \ Z^2 eV$
3. Average kinetic energy of X can be $6.8 \ Z^2 \, eV$
4. Magnetic quantum number of X can have values of $-2, \ -1, \ 0, \ +1, +2$

24. Which of the following properties is true regarding diagonal relationship between Li and Mg ?
    1. Li and Mg salts do not impart color in oxidizing flame
    2. Li or Mg nitrate liberates only one paramagnetic gas on heating
    3. Their bicarbonates are used as antihistamines
    4. Their carbonates decompose on heating to form a gas whose molecule has net zero dipole moment

25. Select the correct statement from the following.
    1. Wrought iron is prepared by oxidizing the impurities of pig iron in a reverberatory furnace lined with Magnesium oxide
    2. Wrought iron is prepared by oxidizing the impurities of cast iron in a reverberatory furnace lined with Haematite.
    3. Wrought iron is prepared by reducing the impurities of cast iron in a blast furnace lined with coke
    4. Wrought iron is prepared by reducing the impurities of pig iron in a blast furnace lined with silica

Section B

26. Which of the following pairs undergoes coupling reaction to form an Azo compound?

    1. 
    2. 
    3. 
    4. Diazotised sulphanilic acid + N, N-Dimethylaniline (in acidic medium)

27. Which of the following forms white crystalline precipitate with Sodium bisulphite?

1. 2,2,4,4-Tetramethyl-3-pentanone
2. Acetaldehyde
3. Benzophenone
4. Glucose

28. Consider the following reaction:

$$\underset{(A)}{Na_2S + Na_2[Fe(CN)_5(NO)]} \rightarrow \underset{(B)}{Violet\ colour\ compound}$$

Identify the correct statement(s) regarding the complexes A and B

1. Effective atomic number of iron in both complexes is same
2. Both complexes exhibit structural isomerism
3. The magnetic moment of the complexes A & B are differ
4. All the atoms of either anion of A or B are on same plane

29. Which of the following is/are correct?

1. Nitric oxide is paramagnetic and depletes ozone layer in upper atmosphere
2. One mole of ozone upon complete decomposition gives 1.5 moles of dioxigen
3. Acidic strength increases from H_2Te to H_2O
4. TeO_2 is stronger oxidizing agent then SO_2 but weaker reducing agent compare to SO_2

30. $Ni_2O_3(s) + H_2O(l) + 2e^- \rightarrow 2NiO(s) + 2OH^-(aq)$
$E^0 = 0.4V$

$FeO(s) + H_2O(l) + 2e^- \rightarrow Fe(s) + 2OH^-(aq)$
$E^0 = -0.87V$

The overall reaction for the spontaneous cell has a value of $\Delta H^0 = -490kJ/mol$
Which of the following option(s) is/are correct?

1. Cell reaction is $Ni_2O_3(s) + Fe(s) \rightarrow 2NiO(s) + FeO(s)$
2. E^0_{cell} is 1.27 V
3. Efficiency of cell is 50.02 %
4. The maximum amount of electrical energy that can be obtained is equal to 245.11 kJ per mole of formation of Nickel(II) oxide

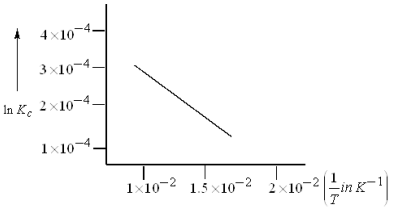
31. K_2MnO_4 In acidic medium, it changes from green to purple after some time. This change is accompanied by

1. Disproportionation of K_2MnO_4
2. Decrease in pH of solution
3. Separation of the brown or black mass from the solution
4. Formation of tetrahedral oxoanion of Mn with oxidation state +7

32. A schematic plot of $\ln K_c$ versus inverse of temperature for reaction is shown below

Which of the following statement(s) is/are correct regarding the reaction?

$xA \rightleftarrows yB \quad K_C = \frac{[B]^y}{[A]^x}$

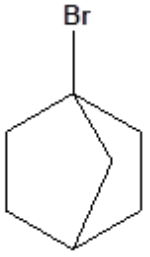


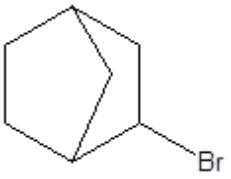
1. Sign of $\Delta_r H^0$ is positive
2. Sign of $\Delta_r S^0$ is negative
3. $\Delta_r G^0$ is negative at all temperature
4. At equilibrium, rate of consumption of A is equal to rate of formation of B

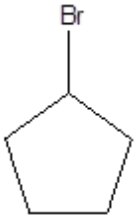
Section C

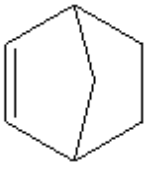
33. *An optically active organic compound P has molecular formula $C_7H_{11}Br$, and it does not declourize bromine water. P on treatment with potassium tertiary butoxide in tertiary butanol yields Q (C_7H_{10}). The compound Q on oxidation with acidified solution of $KMnO_4$ yields R $(C_7H_{10}O_4)$, which does not undergo intramolecular dehydration on heating with P_2O_5. The compound R on heating with NaOH (CaO) gives S (C_5H_{10}). S on heating with dry bromine vapour given $T\ (C_5H_9Br)$ as only monobromo derivative*

Select the incorrect structures

1. P = 

2. P = 

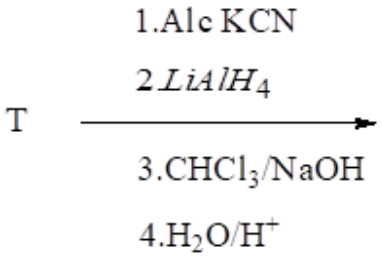
3. T = 

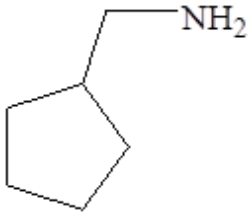
4. Q = 

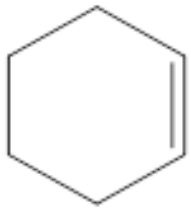
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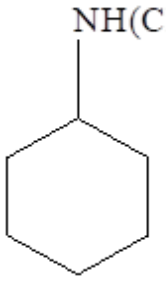
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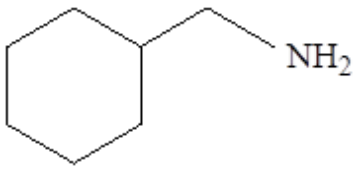
Major product of the following reaction scheme is



1. 

2. 

3. 

4. 

35. *A 30 g of liquid X(M= 60g/mol) is mixed with 40 g of liquid Y (80g/mol) in a closed container at 27^0C.*

Following information is given:

	X (ℓ)	X (g)	Y (ℓ)	Y (g)
$\Delta_f H$ (kJ/mol)	2.00	10.3	10.3	17.96

$\Delta S_{vap}\,(X) = 20.75\ J/mol\ K$

$\Delta S_{vap}\,(Y) = 20.00\ J/mol\ K$

[Given that $e^{5/6} = 2.3,\ e^{2/3} = 2,\ R = 8.3\ J\ mol^{-1}K^{-1}$,
X and Y form ideal mixture for all compositions. ΔH and ΔS values of each liquid remain constant till its boiling point

Select correct statement:

1. Liquid Y is more volatile than liquid X
2. Mass fraction of Y in vapour phase is 0.14 at 27^0C
3. Mass fraction of Y in liquid phase is 0.25 at 27^0C
4. All the above

36. *A 30 g of liquid X(M= 60g/mol) is mixed with 40 g of liquid Y (80g/mol) in a closed container at 27^0C.*

Following information is given:

	X (ℓ)	X (g)	Y (ℓ)	Y (g)
$\Delta_f H$ (kJ/mol)	2.00	10.3	10.3	17.96

$\Delta S_{vap}(X) = 20.75\ J/mol\ K$

$\Delta S_{vap}(Y) = 20.00\ J/mol\ K$

[Given that
$e^{5/6} = 2.3,\ e^{2/3} = 2,\ R = 8.3\ J\ mol^{-1}K^{-1}$,
X and Y form ideal mixture for all compositions. ΔH *and* ΔS *values of each liquid remain constant till its boiling point*
Total pressure of above the liquid mixture at 27^0C is_____atm

1. 0.23
2. 0.53
3. 0.47
4. 0.57

Mathematics

Section A

37. Let the sequence $\{a_i\}_{i=0}^{\infty}$ be defined by $a_0 = \frac{1}{3}$ and $a_n = 1 + ((a_{n-1}) - 1)^2$. The value of the product $\prod_{i=0}^{\infty} a_i$ $(where\ \prod_{i=0}^{\infty} a_i = a_0a_1a_2.....)$ is _____

1. $\frac{1}{5}$
2. $\frac{1}{3}$
3. $\frac{2}{3}$
4. $\frac{3}{5}$

38. Number of ways are there to insert + ‘s (plus signs) between the digits of 111111111111111 (fifteen 1’s) so that the result will be a multiple of 30?

1. 2000
2. 1001
3. 1002
4. 2002

39. A variable point P is on the circle $x^2 + y^2 = 4$ on XY-plane. From point P perpendicular PN is drawn to the line $L : x - 2 = y - 2 = -z$, then minimum length PN drawn is

1. $4\sqrt{2}$
2. $\sqrt{\frac{3-2\sqrt{2}}{6}}$
3. $\frac{1}{3}\sqrt{\frac{3-2\sqrt{2}}{2}}$
4. $\frac{2\sqrt{2}}{\sqrt{3}}\sqrt{\frac{3-2\sqrt{2}}{2}}$

40. If $M = \begin{pmatrix} \frac{5}{2} & \frac{3}{2} \\ -\frac{3}{2} & -\frac{1}{2} \end{pmatrix}$, then which of the following matrices is equal to M^{2024}?

1. $\begin{pmatrix} 3037 & 3036 \\ -3036 & -3035 \end{pmatrix}$
2. $\begin{pmatrix} 3037 & -3036 \\ 3036 & -3035 \end{pmatrix}$
3. $\begin{pmatrix} 3037 & 3036 \\ -3036 & -3036 \end{pmatrix}$
4. $\begin{pmatrix} 3037 & 3035 \\ -3036 & -3035 \end{pmatrix}$

41. Given that x is a positive real, the maximum possible value of

$\sin\left(\tan^{-1}\left(\frac{x}{25}\right)-\tan^{-1}\left(\frac{x}{36}\right)\right)$ is equal to _____

1. $\frac{11}{25}$
2. $\frac{11}{61}$
3. $\frac{60}{61}$
4. $\frac{25}{36}$

42. The value of $\lim\limits_{n\to\infty}\left[\frac{1}{n^2}\sum_{1\le i<j\le n}\tan^{-1}\left(\frac{i}{n}\right)\tan^{-1}\left(\frac{j}{n}\right)\right]$ $(i,j\in integers)$ is

1. $\frac{1}{4}\left(\frac{\pi-\ln 4}{4}\right)^2$
2. $\frac{1}{2}\left(\frac{\pi-\ln 2}{2}\right)^2$
3. $\frac{1}{2}\left(\frac{\pi-\ln 4}{4}\right)^2$
4. $\frac{1}{2}\left(\frac{\pi-\ln 2}{4}\right)^2$

43. Let $x\in\left[-\frac{\pi}{2},-\frac{5\pi}{12}\right]$. Then the maximum value of $y=\tan\left(x+\frac{3\pi}{4}\right)-\tan\left(x+\frac{\pi}{4}\right)+\cos\left(x+\frac{\pi}{4}\right)$ is equal to _____

1. $\frac{12}{5}\sqrt{2}$
2. $\frac{11}{6}\sqrt{2}$
3. $\frac{11}{6}\sqrt{3}$
4. $\frac{12}{5}\sqrt{3}$

Section B

44. Assume that α,β,γ satisfy $0<\alpha<\beta<\gamma<2\pi$, If $\cos(x+\alpha)+\cos(x+\beta)+\cos(x+\gamma)=0$ for all $x\in R$, then which of the following is/are CORRECT ?

1. $\alpha-\beta=\frac{-2\pi}{3}$
2. $\gamma-\beta=\frac{2\pi}{3}$
3. $\alpha-\beta=\frac{-\pi}{3}$
4. $\gamma-\beta=\frac{\pi}{3}$

45. Let A, B, C be angles of a triangle with $\cos^2 A+\cos^2 B+2\sin A\sin B\cos C=\frac{10}{7}$ and $\cos^2 B+\cos^2 C+2\sin B\sin C\cos A=\frac{11}{8}$ then which of the following is/are CORRECT ?

1. $\sin^2 A=\frac{5}{8}$
2. $\sin^2 A=\frac{4}{7}$
3. $\cos^2 C+\cos^2 A+2\sin C\sin A$ $\sin B\ can\ be\ \frac{45+8\sqrt{15}+20\sqrt{3}}{56}$
4. $\cos^2 C+\cos^2 A+2\sin C\sin A$ $\sin B\ can\ be\ \frac{45-8\sqrt{15}-20\sqrt{3}}{56}$

46. An ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ is tangent to each of the circles $(x-2)^2+y^2=4$ and $(x+2)^2+y^2=4$ at two points. If the area of the ellipse is minimized, then which of the following is/are CORRECT ?

1. The value of $\frac{a^2}{b^2}$ is $\frac{1}{3}$
2. Minimum area of the ellipse is $3\sqrt{3}\pi$

3. The value of $\frac{a^2}{b^2}$ is 3

4. Minimum area of the ellipse is $6\sqrt{3}\pi$

47. If $I = \int\limits_{\frac{1}{\sqrt{2}}}^{1} \frac{e^{x^4}}{x} dx$ then which of the following is/are CORRECT ?

1. $I > 2e$
2. $I < \left(\sqrt{2} - 1\right)e$
3. $I > \sqrt{\frac{\sqrt{e}}{6}}$
4. $I < e^{\frac{1}{16}} + e$

48. Let A, B be complex 2×2 matrices satisfying the relation A = AB – BA . Then which of the following is/are CORRECT ?

1. A is a nil potent matrix
2. $|A| = 1$
3. Trace $\left(A^2\right) = 0$
4. A is an orthogonal matrix

49. The number of nonempty subsets $S \subseteq \{-11, -10, -9, -8, ...8, 9, 10, 11\}$ that satisfy $|S| + \min(S).\max(S) = 0$ is R, then R is divisible by ($|S|$ denotes number of elements of set S, min (S) denotes minimum element of set S and max(S) denotes maximum element of set S)

1. 5
2. 11
3. 47
4. 89

50. Considering the equation of the circle $x^2 + y^2 = 9$ and a line $y = 1$ which divides the circle into two regions let small region be R_1 and other be R_2. Which of the following statements is/are CORRECT?

1. Radius of biggest circle lying in region R_1 is 1.
2. Radius of biggest circle lying in region R_2 is 2
3. If two circles of equal radii touching each other externally lying in region R_2 then maximum value of their radii is $2\left(\sqrt{3} - 1\right)$.
4. If the length of intercept made by line $y = 1$ on a circle which is concentric with given circle is $2\sqrt{5}$ then radius of the circle is $\sqrt{6}$.

Section C

51. *Consider on obtuse angled triangle ABC in which the difference between the largest and the smallest angle is $\frac{\pi}{2}$ and whose sides are in arithmetic progression. Suppose that the vertices of this triangle lie on a circle of radius 1.*

Let a be the length of the altitude drawn from the vertex containing

largest angle of ΔABC. Then the value of $(8a-7)^2$ is

1. 5
2. 7
3. 9
4. 11

52. *Consider on obtuse angled triangle ABC in which the difference between the largest and the smallest angle is* $\frac{\pi}{2}$ *and whose sides are in arithmetic progression. Suppose that the vertices of this triangle lie on a circle of radius 1.*

The square of in-radius of the ΔABC is

1. $\frac{1}{2}$
2. $\frac{1}{4}$
3. $\frac{1}{8}$
4. $\frac{1}{16}$

53. *The polynomials are the expression written in the form of* $a_n x^n + a_{n-1}x^{n-1} + ... + a_1 x + a_0$ *. The values of the variable which set the value of entire polynomial to zero are known as roots of the polynomial or zeros of polynomial*

For certain real numbers a, b and c, the polynomial $g(x) = x^3 + ax^2 + x + 10$ has three distinct roots, and each root of g(x) is also a root of the polynomial $f(x) = x^4 + x^3 + bx^2 + 100x + c$ .The value of f(1) is

1. –7007
2. 7007
3. 4001
4. -4001

54. *The polynomials are the expression written in the form of* $a_n x^n + a_{n-1}x^{n-1} + ... + a_1 x + a_0$ *. The values of the variable which set the value of entire polynomial to zero are known as roots of the polynomial or zeros of polynomial*

Suppose a, b denote the distinct real roots of the quadratic polynomial $x^2 + 10x - 1010$ and suppose c, d denote the distinct complex roots of the quadratic polynomial $x^2 - 10x + 1010$. Then the value of ac(a–c)+ad(a–d)+bc(b–c)+bd(b–d) is

1. 0
2. 1000
3. 1010
4. 2000